# DONGHAO HUANG

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Singapore Citizen | Patents: 10 granted, 50+ pending

## PROFESSIONAL SUMMARY

Technology executive with 26+ years of experience across software engineering, Web3, AI/ML/GenAI, and payments innovation. Currently leading global R&D for Mastercard Emerging Technology (AI/Web3) while pursuing a Doctor of Engineering in Generative AI. Author of 24 published research works spanning LLM deployment, LLM/ML model optimization and applications, and multi-agent systems. Proven track record of translating academic research into enterprise-scale financial services solutions.

## EDUCATION

Aug 2023 – May 2026 (ongoing) Singapore Management University Singapore
**Doctor of Engineering (EngD) in Generative AI**

- Awarded IMDA SG Digital Scholarship (Postgraduate) 2025
- Research focus: Retrieval-Augmented Generation (RAG), Edge AI, LLM and ML model optimization, multi-agent systems, sentiment analysis, reasoning LLMs
- Published 24 peer-reviewed papers at top AI conferences and journals (AAAI, AAMAS, NAACL, IJCNN, PAKDD, IEEE CAI, IEEE SSCI, IEEE ICDM, IEEE Intelligent Systems, Artificial Intelligence Reviews)

Jan 2004 – Jan 2008 National University of Singapore Singapore
**Master of Technology in Software Engineering**

Sep 2001 – Aug 2002 Centre for Optoelectronics, NUS Singapore
**Certificate, EDB Research and Training Programme**

Sep 1998 – Jul 2001 Peking University Beijing, China
**Master of Science in Theoretical Physics**

- Published 2 peer-reviewed papers in physics journals

Sep 1989 – Jul 1995 Tsinghua University Beijing, China
**Bachelor of Engineering in Engineering Physics**

## PROFESSIONAL EXPERIENCE

Aug 2025 – to date Mastercard (U.S.) Arlington, VA, USA
**VP, R&D / Global EmTech Lead**

- Lead global R&D initiatives in AI Emerging Technology (EmTech) domains
- Lead EmTech research for 5 strategic "Big Bets" projects – high-impact, confidential initiatives aimed at shaping Mastercard's next-generation payment infrastructure and AI-powered commerce solutions
- Organized and hosted SENTIRE'25 (15th Workshop on Sentiment Analysis and Linguistic Linked Data) co-located with IEEE ICDM 2025, November 13th, Washington DC.

May 2022 – Jul 2025 Mastercard Asia/Pacific Pte Ltd Singapore

**VP, R&D / Global EmTech Lead**

- Lead global R&D initiatives in Emerging Technology (EmTech) domains: AI / Machine Learning / Generative AI and Web3.
- **AI/LLM/GenAI Initiatives:**
  - Led EmTech research for Mastercard Assistant Platform (MAP)**,** a suite of digital assistants designed to simplify the customer onboarding process for Mastercard products
    - Built the initial Retrieval-Augmented Generation (RAG) prototype using LangChain, FAISS vector databases, and HuggingFace embeddings to power enterprise-grade knowledge systems.
    - Conducted early benchmarking of open-source LLMs (Llama 2, Orca 2) on NVIDIA A40 and L40 GPUs (48GB each), demonstrating 24-49% inference speed improvement on L40 vs A40.
    - Developed a novel Repetition-aware Performance (RAP) evaluation framework for LLM quality assessment, now fully deployed in the MAP production environment.
    - Led research on edge-aware LLM deployment using the Ollama framework across diverse hardware platforms (NVIDIA RTX A6000, NVIDIA RTX 4090, NVIDIA Jetson AGX Orin, Apple M-series—directly informing MAP's deployment strategy.
  - Pioneered enterprise GenAI applications using LLMs (GPT-4/5, Claude, Llama, Qwen) for payments/commerce automation
    - Fine-tuned 70B+ parameter models (Llama-3.1-70B, Qwen2-72B) using QLoRA on multi-GPU setups (4× NVIDIA L40 GPUs) for Chinese-English translation and logical reasoning
    - Developed multi-agent systems for invoice reconciliation achieving 99.9% accuracy using edge-deployed LLMs (qwen2.5:32b)
    - Developed hierarchy multi-agent systems for conversational payment agent (HMASP) - first LLM-based multi-agent system to implement end-to-end agentic payment workflows, enabling external AI agents to converse and complete payments using natural language with minimal integration overhead
    - Developed CMAS4G2, a cooperative multi-agent system for automated Gherkin acceptance test generation using Microsoft's AutoGen Swarm framework; demonstrated open-weight models on consumer hardware (NVIDIA RTX 4090, 24GB VRAM) approach proprietary performance (GPT-OSS 20B: 87.6% success vs Claude Sonnet 4: 90.4%)
    - Created LLM-as-a-Judge (LAJ) evaluation framework achieving 47× cost reduction vs GPT-5 with negligible accuracy loss for automated test coverage assessment
    - Evaluated reasoning LLMs (DeepSeek-R1, Qwen3, GPT-OSS) for sentiment analysis with cross-platform performance benchmarking on NVIDIA H100, NVIDIA GeForce RTX 4090, Apple M3 Max
    - Developed energy efficiency metrics (Energy per Inference, Performance-to-Power Ratio) for sustainable AI deployment
- **Web3 Initiatives:**
  - Developed Mastercard Tokenized Card NFT solution, enabling seamless tokenization of payment cards as Soul Bound Tokens (SBTs) into Web3 wallets
    - Enables users to store credit cards securely as non-transferable NFTs, providing publicly verifiable digital identity for web3 wallet addresses

- Integrates with crypto exchanges, merchant partners, and payment acquirers through smart contract architecture
- Simplifies Web3 user onboarding by eliminating the need to switch between multiple applications and cumbersome digital wallet funding processes

- Developed DigiPay, a Web3 micropayment stablecoin solution leveraging Layer 2 blockchain and ERC-4337 account abstraction
  - Implemented Digicents smart contract token with high TPS, low-latency blockchain for fast, efficient, and affordable transactions
  - Built smart contract wallet accounts with auto-top-up and auto-withdrawal features, eliminating manual on-ramp/off-ramp processes
  - Integrated with Tokenized Card NFT for secure funding and transaction authentication within the Digicents ecosystem

Aug 2011 – May 2022 Mastercard Asia/Pacific Pte Ltd Singapore

**VP, R&D / Head of Singapore R&D**

- Found and lead Singapore R&D team in Mastercard Foundry (previously known as Mastercard Labs) – Mastercard's global innovation and R&D division; managing a high performing R&D team based in Singapore.
- Work on innovation and R&D in payment/commerce domains across lots of latest and cutting-edge technologies, including but not limited to, EMV Chip technologies, Internet of Things, QR Payments, Connected Car, Humanoid Robots, Conversational AI, AR/VR, Blockchain and Digital Assets, Web3, NFT, DeFi, Metaverse and so on
- Drive complex initiatives from idea through to commercialization hand-over; facilitate the transition of initiatives into Mastercard commercialization process
- As a champion for Technological Innovation, collaborate with various partners (internal departments, external partners, universities) to ensure Mastercard attains a reputation for technological excellence in the Payments space
- **Key Commercialized Projects**
  - Nov 2021: Mastercard partners with city of New Orleans and MoCaFi to announce Crescent City Card program: https://www.mastercard.com/news/press/2021/november/mastercard-partners-with-city-of-new-orleans-and-mocafi-to-announce-crescent-city-card-program/
  - Sep 2020: Mastercard partners with Samsung, Airtel and Asante to drive digital inclusion in Africa through Pay-on-Demand services: https://newsroom.mastercard.com/mea/press-releases/mastercard-partners-with-samsung-airtel-and-asante-to-drive-digital-inclusion-in-africa-through-pay-on-demand-services/
  - Oct 2018: Mastercard, Centenary Bank and M-KOPA Solar roll-out first of its kind energy solution in Uganda: https://newsroom.mastercard.com/mea/press-releases/mastercard-centenary-bank-and-m-kopa-solar-roll-out-first-of-its-kind-energy-solution-in-uganda/
  - Feb 2018: Mastercard Uses Facebook Messenger To Help Small Businesses Go Digital: https://newsroom.mastercard.com/press-releases/mastercard-uses-facebook-messenger-help-small-businesses-go-digital/
  - Sep 2017: SPH Buzz Partners With Mastercard To Launch New Hybrid Store Concept: https://newsroom.mastercard.com/asia-pacific/press-releases/sph-buzz-partners-with-mastercard-to-launch-new-hybrid-store-concept/

- Feb 2017: BharatQR – World's first interoperable QR code acceptance solution launched in India: https://newsroom.mastercard.com/asia-pacific/press-releases/bharatqr-worlds-first-interoperable-qr-code-acceptance-solution-launched-in-india/
- May 2016: MasterCard Powers First Commerce Application within SoftBank Robotics' Humanoid Robot "Pepper": https://newsroom.mastercard.com/press-releases/mastercard-powers-first-commerce-application-within-softbank-robotics-humanoid-robot-pepper/
- Sep 2015: MasterCard Transforms Aid Distribution: https://newsroom.mastercard.com/press-releases/mastercard-transforms-aid-distribution/
- Aug 2012: StarHub unveils 'SmartWallet', in collaboration with DBS, EZ-Link and MasterCard: https://www.starhub.com/about-us/newsroom/2012/august/starhub-unveils-smartwallet--in-collaboration-with-dbs--ez-link-.html

Jul 2008 – Aug 2011 PayPal Singapore Development Centre Singapore

**Senior Software Engineer / Team Lead**

- Lead a team of software developers, working on a pipeline of projects related to Brazil to cater to this fast growing market
- Subject Matter Expert in limits / verification / compliance domain
- **Key Commercialized Projects**
  - Led 12-developer team delivering largest EU compliance project (EU KYB Receiving Limit) in 8 months
  - Built China UnionPay (CUP) integration enabling real-time transaction processing

Oct 2007 – Jul 2008 Streaming21 (Singapore) Pte Ltd Singapore

**Software R&D Manager**

- Established an R&D centre and led development of Media Relay Server, a leading IPTV streaming server.

Jan 2006 – Sep 2007 Schmidt Electronics (S.E.A.) Pte Ltd Singapore

**Leader, Software Development**

Feb 2005 – Dec 2005 Unaxis Singapore Pte Ltd Singapore

**Software Engineer**

Sep 2002 – Feb 2005 ASM Technology Singapore Pte Ltd Singapore

**Software Engineer**

Aug 1995 – Aug 1998 Beijing Zhida Technology Pte Ltd Beijing, China

**Software Engineer**

## VOLUNTEERING EXPERIENCE

Nov 2025 – to date Bao Yan (Singapore) Education Society

**President**

Jun 2016 – Jun 2020 Peking University Alumni Association (Singapore)

**Vice President**

Jan 2015 – Jan 2019 Startupbootcamp FinTech Singapore
**Mentor**

Mar 2011 – Dec 2015 ConneXions International Singapore
**IT and Payment Consultant**

Nov 2006 – Jul 2008 St. Andrew's Autism Centre Singapore
**Technology Consultant**

## TECHNICAL SKILLS

- **AI/ML/GenAI:** LLMs (GPT-4, GPT-5, Claude, Gemini, Llama, Qwen, DeepSeek-R1, Mistral, Magistral, Phi, Nemotron), PyTorch, Transformers, LLM/ML Model Fine-tuning and Optimization, PEFT, LoRA/QLoRA Fine-tuning, LlamaFactory, Unsloth, RAG, LangChain, LangGraph, Autogen, Ollama, LM Studio, Vector Databases, FAISS, Chroma, Prompt Engineering, Few-shot Learning, NLP, Sentiment Analysis, Agentic AI, OpenClaw, NemoClaw, Edge AI Deployment, Energy-efficient Inference
- **GPU/Hardware:** NVIDIA H100, L40, A40, RTX 4080, RTX 4090, RTX A6000, Jetson AGX Orin, DGX Spark; Apple M-series
- **Languages & Frameworks:** Python, C/C++, Java, NodeJS, React, Vue, Spring Boot/Cloud, Solidity
- **Cloud & Infrastructure:** AWS, Azure, Heroku; Microservices, CI/CD, Docker
- **Web3/Blockchain:** Blockchain, Smart Contract, IPFS, NFT, DeFi

## PATENTS

(10 granted, 50+ pending — all assigned to Mastercard)

**Granted Patents**

- **US12417452B2** – Smart chip payment acceptance (2025)
- **US12217246B2** – Method and system for use of an EMV card in a multi-signature wallet for cryptocurrency transactions (2025)
- **US11715100B2** – Electronic system and computerized method for verification of transacting parties to process transactions (2023)
- **US11615406B2** – Method and system for providing a service at a self-service machine (2023)
- **US10535034B2** – Item delivery management systems and methods (2020)
- **US10482499B2** – Method for conducting a transaction (2019)
- **US10423949B2** – Vending machine transactions (2019)
- **US10083427B2** – Method for receiving an electronic receipt of an electronic payment transaction into a mobile device (2018)
- **US9836729B2** – Method and system for conducting a payment transaction and corresponding devices (2017)
- **US8712914B2** – Method and system for facilitating micropayments in a financial transaction system (2014)

**Selected Pending Applications**

- **WO2026063867A1** – Method and system for seamless cross-network physical-digital (phy-gital) experience (2024)
- **US20260080407A1** – Method and system for seamless cross-network physical-digital (phy-gital) experience (2024)
- **WO2024058723A1** – Digitization of payment cards for web 3.0 and metaverse transactions (2022)
- **US20230059546A1** – Access control system (2021)
- **EP4356332A2** – Method and system for mediated cross ledger stable coin atomic swaps using hashlocks (2021)
- **US20210406887A1** – Method and system for merchant acceptance of cryptocurrency via payment rails (2020)

- **US20210158316A1** – Electronic system and computerized method for controlling operation of service devices (2019)
- **US20200175489A1** – Method and system for QR code originated vending (2017)
- **WO2017200643A1** – Method and system for allocating and transacting with multiple non-financial commodities (2016)
- **US20180025332A1** – Transaction facilitation (2016)

*… and 40+ additional pending patent applications across payment technology, blockchain, AI, and IoT domains.*

## RESEARCH PAPERS

(papers at top-tier venues are highlighted in yellow; full texts are accessible via the hyperlinked paper titles)

**Artificial Intelligence**

1. Chua, J., Huang, D. & Wang, Z. (2026). A Novel Hierarchical Multi-Agent System for Payments Using LLMs. Accepted for publication in *The 30th Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD 2026)*.
2. Huang, D., & Wang, Z. (2026). Task Complexity Matters: An Empirical Study of Reasoning in LLMs for Sentiment Analysis. Accepted for publication in *The 30th Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD 2026)*.
3. Huang, D., Chua, J. K., & Wang, Z. (2026). Beyond Task Success: Measuring Workflow Fidelity in LLM-Based Agentic Payment Systems. Accepted for publication in *Trends and Applications in Knowledge Discovery and Data Mining. PAKDD 2026*.
4. Huang, D., Pai, P., & Wang, Z. (2026). Entity Matching with LLMs at Scale: Accuracy, Cost, and Reasoning Effects. Accepted for publication in *Trends and Applications in Knowledge Discovery and Data Mining. PAKDD 2026*.
5. Huang, D., Chew, S., & Wang, Z. (2026). CMAS4G2: A Locally Deployed Cooperative Multi-Agent System for Gherkin Acceptance Test Generation. Accepted for publication in *Trends and Applications in Knowledge Discovery and Data Mining. PAKDD 2026*.
6. Huang, D. & Pandey, M. (2026). An Agentic Voice-Based Assistant for Interactive Conversation and Guidance in Real-World Environments. Accepted for publication in Demonstration Track, *Proc. of the 25th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2026)*, Paphos, Cyprus, May 25–29, 2026.
7. Chua, J. K., Tse, C. H., Huang, D., & Wang, Z. (2026). ConvPayMAS: Conversational Payment Multi-Agent System with Agent-to-Agent Protocol and Three-Mandate Verification. Accepted for publication in Demonstration Track, *Proc. of the 25th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2026)*, Paphos, Cyprus, May 25–29, 2026.
8. Huang, D., Malwe, G., & Wang, Z. (2026). When Agents Fail to Act: A Diagnostic Framework for Tool Invocation Reliability in Multi-Agent LLM Systems. Accepted for publication in *2026 The 9th International Conference on Artificial Intelligence and Big Data (ICAIBD 2026)*.
9. Huang, D., Chew S., Dutkiewicz A., & Wang, Z. (2026). LLM-as-a-Judge for Scalable Test Coverage Evaluation: Accuracy, Operational Reliability, and Cost. Presented at *AAAI 2026 Workshop on Next-Gen Code Development with Collaborative AI Agents*.
10. D. Huang, N. Teo, A. Chekuru, E. Deutschman and Z. Wang, "Deploying Reasoning LLMs for Sentiment Analysis: Architecture Trade-offs on Consumer Hardware," 2025 IEEE International Conference on Data Mining Workshops (ICDMW), Washington, DC, USA, 2025, pp. 2048-2055, doi: 10.1109/ICDMW69685.2025.00249.
11. S. Samuel, D. Huang, J. T. Y. Hong and Z. Wang, "Integrating LLM Sentiment Analysis into Machine Learning for Soccer Betting," 2025 IEEE International Conference on Data Mining Workshops (ICDMW), Washington, DC, USA, 2025, pp. 2156-2164, doi: 10.1109/ICDMW69685.2025.00262.
12. D. Huang and Z. Wang, "Explainable Sentiment Analysis With DeepSeek-R1: Performance, Efficiency, and Few-Shot Learning," in *IEEE Intelligent Systems*, vol. 40, no. 6, pp. 52-63, Nov.-Dec. 2025, doi: 10.1109/MIS.2025.3614967.
13. Huang, D., Ng, D., Pen, H., Wang, Z., & Cambria E. (2025). Evaluating the Impact of LLM-Manipulated Content on Fake News Detection. In: Yuan, S., Malliaros, F., Zheng, X. (eds) *Trends and Applications in Knowledge*

*Discovery and Data Mining. PAKDD 2025*. Lecture Notes in Computer Science(), vol 15835. Springer, Singapore. https://doi.org/10.1007/978-981-96-8197-6_28

14. Teo, N., Huang, D., Cambria, E., Wang, Z. (2025). Large Language Models for Logical Fallacy Detection. In: Yuan, S., Malliaros, F., Zheng, X. (eds) *Trends and Applications in Knowledge Discovery and Data Mining. PAKDD 2025*. Lecture Notes in Computer Science(), vol 15835. Springer, Singapore. https://doi.org/10.1007/978-981-96-8197-6_29
15. D. Huang and Z. Wang, "LLMs at the Edge: Performance and Efficiency Evaluation with Ollama on Diverse Hardware," *2025 International Joint Conference on Neural Networks (IJCNN)*, Rome, Italy, 2025, pp. 1-8, doi: 10.1109/IJCNN64981.2025.11228317.
16. Huang, D., Nguyen, T. S., Liausvia, F., & Wang, Z. (2025, April). RAP: A Metric for Balancing Repetition and Performance in Open-Source Large Language Models. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)* (pp. 1479-1496).
17. Huang, D., Malwe, G., Varshney, R., Tse, A., & Wang, Z. (2025, March). Scalable Invoice Reconciliation for SMEs: Edge-Deployed LLMs in Multi-Agent Systems. In *2025 International Conference on Data Science, Agents & Artificial Intelligence (ICDSAAI)* (pp. 1-6). IEEE.
18. D. Huang and Z. Wang, "Optimizing Chinese-to-English Translation Using Large Language Models, " *2025 IEEE Symposium on Computational Intelligence in Natural Language Processing and Social Media (CI-NLPSoMe Companion)*, Trondheim, Norway, 2025, pp. 1-7, doi: 10.1109/CI-NLPSoMe64976.2025.10970768.
19. D. Huang and Z. Wang, "Logical Reasoning with LLMs via Few-Shot Prompting and Fine-Tuning: A Case Study on Turtle Soup Puzzles," *2025 IEEE Symposium on Computational Intelligence in Natural Language Processing and Social Media (CI-NLPSoMe Companion)*, Trondheim, Norway, 2025, pp. 1-5, doi: 10.1109/CI-NLPSoMeCompanion65206.2025.10977921.
20. Wang, Z., Huang, D., Cui, J. et al. (2025). A review of Chinese sentiment analysis: subjects, methods, and trends. Artif Intell Rev 58, 75 (2025).
21. Huang, D., Fu, X., Yin, X., Pen, H., & Wang, Z. (2024). Automating maritime risk data collection and identification leveraging large language models. In *2024 IEEE International Conference on Data Mining Workshops (ICDMW)* (pp. 433–439). IEEE.
22. Huang, D., Hu, Z., & Wang, Z. (2024, June). Performance Analysis of Llama 2 Among Other LLMs. In *2024 IEEE Conference on Artificial Intelligence (CAI)* (pp. 1081-1085). IEEE.
23. Huang, D., & Wang, Z. (2024, April). Evaluation of Orca 2 Against Other LLMs for Retrieval Augmented Generation. In: Wang, Z., Tan, C.W. (eds) *Trends and Applications in Knowledge Discovery and Data Mining. PAKDD 2024*. Lecture Notes in Computer Science(), vol 14658. Springer, Singapore. https://doi.org/10.1007/978-981-97-2650-9_1
24. Huang, D., Samuel, S., Hyunh, Q. T., & Wang, Z. (2024, April). From Tweets to Token Sales: Assessing ICO Success Through Social Media Sentiments. In: Wang, Z., Tan, C.W. (eds) *Trends and Applications in Knowledge Discovery and Data Mining. PAKDD 2024*. Lecture Notes in Computer Science(), vol 14658. Springer, Singapore. https://doi.org/10.1007/978-981-97-2650-9_5

**Theoretical Physics**

25. Lin, W. B., Huang, D. H., Zhang, X., & Brandenberger, R. (2001). Nonthermal production of weakly interacting massive particles and the subgalactic structure of the universe. *Physical Review Letters*, *86*(6), 954.
26. Huang, D. H., Lin, W. B., & Zhang, X. M. (2000). Remark on approximation in the calculation of the primordial spectrum generated during inflation. *Physical Review D*, *62*(8), 087302.